

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 1

Year: 2020

Subject: 392153 Chemistry III

Section: 15-18

Date: 1 September 2020

Time: 10.00-12.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 2 sections. There will be 8 pages (including this page), 7 questions and they are worth 80 marks (40 points).

วิทยาลัยเทคโนโลยีอุตสาหกรรม

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test papers.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can't be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Part 1: Total score 40 marks = 20%

1. Answer questions about metallic bonding (15 marks)

1.1 How are metallic bonds different from ionic and covalent bond? (3 marks)

.....

.....

.....

.....

.....

.....

1.2 What is the metallic lattice (electron sea model)? (3 marks)

.....

.....

.....

.....

.....

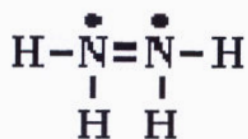
.....

วิทยาลัยเทคโนโลยีอุตสาหกรรม

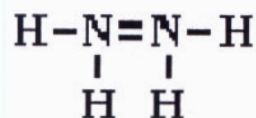
1.3 Give the reason in the table below (9 marks)

Properties of metal materials	Reason
High melting and boiling point	
Good conductor of electricity	
Malleable and ductile	

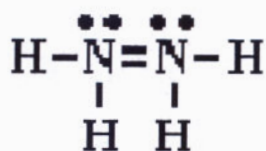
2. Which of following is an acceptable Lewis structure for NH_2NH_2 ? Please explain with scientific reasons. (10 marks)



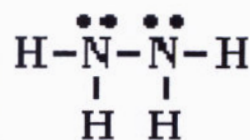
Structure 1



Structure 2



Structure 3



Structure 4

วิทยาลัยเทคโนโลยีอุตสาหกรรม

3. Describe limitations of the octet theory (5 marks)

ศูนย์เทคโนโลยีอุตสาหกรรม

4. Ethene gas (C_2H_4) reacts with bromine gas (Br_2) to produce 1,2-dibromoethane ($\text{C}_2\text{H}_4\text{Br}_2$). Calculate the enthalpy change (ΔH°) of the reaction, given the average bond enthalpies in the table below. (10 marks)

Bond	Average bond enthalpy/kJ mol ⁻¹
Br-Br	193
C-C	346
C=C	614
C-Br	285
C-H	414

[illegible]

.....

.....

.....

.....

.....

.....

.....

Asst.Prof.Dr. Rungsima Yeetsorn

Part 2: Total score 40 marks = 20%

1. Answer the following questions. (20 marks)
- 1.1) Complete the table below.

Formula	Write Lewis structure and calculate the formal charge on each atom	Molecular geometry	Polar/ Nonpolar
NH ₃			
BF ₃			
CH ₂ Cl ₂			
BrF ₄ ⁻			

This image shows a blank sheet of white paper with horizontal blue ruling lines. A single vertical red margin line runs down the right side of the page. The paper is otherwise empty of any text or markings.

Given:	First ionization energy of Al	=	500 kJ/mol
	Second ionization energy of Al	=	1,500 kJ/mol
	Third ionization energy of Al	=	3,000 kJ/mol
	Dissociation energy of F ₂	=	160 kJ/mol
	Electron affinity of F	=	-330 kJ/mol
	Sublimation energy of Al	=	a kJ/mol
	Lattice energy of Aluminium fluoride	=	-5,250 kJ/mol

This image shows a blank sheet of white paper designed for handwriting practice. It features ten horizontal rows, each defined by two parallel dashed black lines. A single solid light-blue vertical line runs down the right side of the page, creating a margin. The paper is otherwise empty of any text or markings.

2.2) Write a flow chart of the Born-Haber cycle of Aluminium fluoride (AlF_3).

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

2.3) Write five steps of Lattice energy of Aluminium fluoride (AlF_3) and calculate each energy in the Born-Haber cycle.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

วิทยาลัยเทคโนโลยีอุตสาหกรรม

3. Why does NH_3 molecule can be dissolved in water, but CCl_4 molecule does not. And, why boiling point of NH_3 molecule is higher than CCl_4 molecule. Please explain your reasons. (5 marks)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Periodic Table of the Elements

Periodic Table of the Elements

1																	18		
1	H																	2	He
																		4.00	Helium
3	Li																	10	Ne
																		6.94	Lithium
4	Be																	20.18	Neon
																		9.01	Beryllium
11	Na																	18	Ar
																		22.99	Sodium
12	Mg																	39.95	Argon
																		24.31	Magnesium
19	K																	36	Kr
																		39.10	Potassium
20	Ca																	84.80	Krypton
																		40.08	Calcium
37	Rb																	54	Xe
																		85.47	Rubidium
38	Sr																	131.29	Xenon
																		87.62	Strontium
55	Cs																	86	Rn
																		132.91	Cesium
56	Ba																	222.02	Radon
																		137.33	Barium
87	Fr																	118	Og
																		223.02	Francium
88	Ra																	294	Oganesson
																		226.03	Radium